

Growth and Growth Disorders

Normal growth: terminology · growth charts · physiology & GH · puberty | Disorders: short stature · GH deficiency · tall stature · abnormal head growth
Yellow = stressed by the doctor · **Blue** = other high-yield · **Red underline** = clue that points to the diagnosis

1. Normal Growth: Terminology & Growth Charts

1.1 Terminology

Term	Definition
Chronological age	Calendar (real) age since birth
Bone (skeletal) age	Age of bone development vs standard → X-ray of wrist or knee compared with standard bone age (e.g. 7-y-old → compare with standard of 7 y)
Short stature	Height < 3rd percentile
Tall stature	Height > 97th percentile
Underweight	Weight < 3rd percentile
Overweight	Weight > 97th percentile

- Length/height is always a **range** (e.g. 100–120 cm) → the range comes from the growth chart

1.2 Saudi vs International Charts

- Saudi charts (2005, El-Mouzan et al.):** healthy Saudi children 0–19 y, all 13 regions, WHO methods → recommended to **replace older / foreign charts**
- Saudi child → **use the Saudi growth chart**
- CDC 2000 vs Saudi: major differences → CDC lower percentiles shifted **up**, upper percentiles overlap / shifted **down** (esp. weight, weight-for-height, BMI)
- Using CDC for Saudi children → ↑ apparent undernutrition, stunting, wasting → **unnecessary referrals, tests, parental anxiety**; ↑ overweight/obesity is alarming
- Al-Nuaim: chart for school-aged Saudi boys → international (NCHS) 50th > Saudi 50th for weight; for height, NCHS 50th ≈ Saudi 95–97th → NCHS **not used anymore**

1.3 Growth

- Normal growth = final common pathway of: **endocrine** (GH) · **environmental** · **nutritional** (enough diet) · **genetic**
- Parental concern about psychological effects of abnormal stature → often why families seek care

1.4 Growth Charts

- Most famous: **NCHS** (National Center for Health Statistics) charts → based on **American** children
- Types: **below 2 y** · **above 2 y** · local (Saudi) · WHO · international
- Boys → **blue** charts; girls → **pink**
- Plot: age on x-axis → read stature (height) and weight percentiles
- Examples: 10-y boy 150 cm → almost 97th; his weight below 50th · 10-y girl 40 kg → ~90th → normal range
- Don't label a child short or tall until at least 2 visits**

Head circumference (birth–36 mo chart)

- After 2–3 y → head circumference becomes nearly **constant** in a normal child
- HC **< 3rd percentile** → **microcephaly**; **> 97th** → **macrocephaly**

1.5 Reading a Chart: Growth Failure

Age	Child A	Child B
6 y	50th	50th
7 y	50th	25th
8 y		10th
9 y		5th
Meaning	Normal	Growth failure

- Growth failure = slow growth rate regardless of stature → crossing > 2 percentiles**

2. Physiology of Growth & Growth Hormone

2.1 Age Stages & Normal Numbers

Stage	Age
Neonate	Birth → 1 month
Infancy	1 month → end of 1st year
Child	1 → 10 y
Adolescent	> 10 y

- Birth: length ~50 cm, weight 3–4.5 kg
- 1st year: 50 → 75 cm → +25 cm
- By 2 y: height velocity ↓ to ~8–10 cm/y
- Growth stops at ~18–19 y

2.2 Infancy

- Linear growth initially **rapid** (up to 1 y)
- Main regulator: **nutritional status**
- If GH = 0 but nutrition is good → infant still grows well

2.3 Childhood

- Pattern starts ~6 mo, predominates from 3 y
- **Hormones become the main regulator**; nutrition less important
- Normal **thyroid** status is required for normal growth
- Height velocity **4–8 cm/year**

2.4 Puberty (Growth)

- Adolescent growth spurt (AGS) → ↑ **androgen** (males) / **estrogen** (females)
- Females: AGS starts **~2 y earlier** than males
- Peak height velocity higher in males: **~10 cm/y (M) vs 8 cm/y (F)**
- Puberty = end of growth

2.5 GH Axis

Hypothalamus: GHRH (+) · somatostatin/SRIF (– negative control)

→ **Anterior pituitary** → **GH** (pulsatile)

▶ GH binds albumin in circulation → goes to **liver**

→ Liver makes **IGF-1** + IGF-binding protein

▶ IGF-1 binds albumin → peripheral tissue (bone, muscle) → binds IGF receptors

→ **Femur (bone) growth + muscle growth**

→ IGF-1 feeds back: ↓ GHRH, ↑ somatostatin

- **Damage at any level** (hypothalamus, pituitary, liver, receptor, bone, muscle, peripheral tissue) → **child will not grow**

Other hypothalamic–pituitary axes (same diagram)

Hypothalamus	Pituitary	Effect
CRH	ACTH	Cortisol
Dopamine (–)	Prolactin	Lactation
GnRH	FSH, LH	Sex steroids, inhibin, gametogenesis
TRH	TSH	T4

2.6 Height Curves (Tanner)

- Height velocity: very high at birth → falls → plateau ~5–6 cm/y in childhood → pubertal peak
- Girls peak earlier (~12 y) and lower; boys later (~14 y) and higher → boys end taller

3. Puberty & Tanner Staging

3.1 Puberty Overview

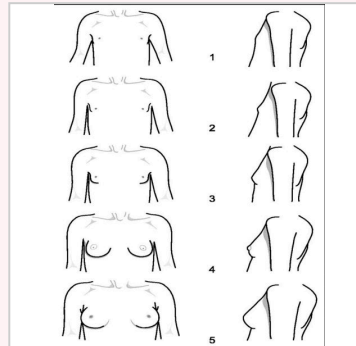
- Hallmark of puberty = **variability**
- Affected by genetic, general health, nutritional, environmental, socioeconomic factors

Female	Male
1) Breast development = first sign	1) Testicular enlargement = first sign
2) Pubic hair	2) Pubic hair
3) Menarche	3) Height spurt

- Pubertal stage → use the **first sign** (breast / testes) together with **pubic hair**

3.2 Breast Stages (Female)

1	Prepubertal: elevation of papilla only
2	"Breast bud" below the areola
3	Further enlargement + elevation of breast and areola
4	Areola forms a secondary mound above breast contour
5	Adult: mound recedes, smooth contour



3.3 Genital Stages (Male)

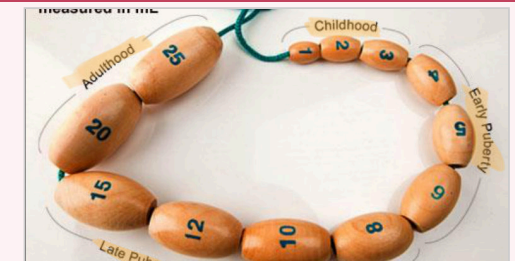
1	Prepubertal genitalia
2	Testes + scrotum enlarge , scrotum reddens and thins, no penile enlargement
3	Penis enlarges: length first , then diameter; testes/scrotum keep growing
4	Further growth + penis lengthens, glans enlarges
5	Adult size and proportion

3.4 Pubic Hair Stages (Both)

1	Fine vellus hair, same as abdominal wall
2	Sparse straight hair at base of penis / along labia
3	More, darker, curlier
4	Adult-type hair, smaller area than adult
5	Spreads to medial thighs; adult male/female pattern

3.5 Orchidometer

- Beads of testicular volume (mL) → **used to assess puberty in boys**
- Childhood 1–3 mL
- Early puberty from ~4 mL
- Mid ~8–10 · late ~12–15
- Adult 20–25 mL



4. Short Stature

4.1 Definitions

- **Short stature**: not a disease → height **2 SD below mean** ($\approx$ 3rd percentile) for sex, chronological age, ideally same ethnic group
- **Growth failure**: slow growth rate regardless of stature → crossing > 2 percentiles

4.2 Proportion

- **Proportionate** vs **disproportionate** → measure **arm span + upper/lower segment**
- **Arm span (AS)** = distance between the 2 middle fingertips → normally **AS = height**
- **U/L ratio** ≈ 1.7 (birth) $\cdot 1.3$ (3 y) $\cdot 1.0$ (14 y) $\cdot 0.9$ (adult)

4.3 Target (Mid-Parental) Height

- Used for short children (FH = father's height, MH = mother's height, cm)
- **Boy**: $[FH + (MH + 13)] / 2 \pm 5$
- **Girl**: $[(FH - 13) + MH] / 2 \pm 5$

4.4 Constitutional Delay vs Familial Short Stature

	Constitutional delay (CDGP)	Familial short stature
Nature	"Temporary delay"	Normal variant
Course	Normal birth size → ↓ height velocity in first 3–5 y → ↓ height percentile	Height < -2 SD, low-normal velocity
Bone age	Delayed = hallmark	Normal
Final height	Complete catch-up	Reaches mid-parental height
Other	Family history	Normal labs; family may carry genetic variants

4.5 Syndromes

- **Down syndrome** → short by nature; **have their own growth chart**
- **Turner syndrome** → **one X chromosome**
 - ▶ Can be short stature while otherwise completely normal
 - ▶ Some: **wide neck**, **widely spaced nipples**, congenital cardiac disease; intelligence good
- **Sanjad-Sakati syndrome** → **rare worldwide, common in the Middle East**
 - ▶ Autosomal recessive (consanguinity) → defective parathyroid development → **hypoparathyroidism** → ↓ Ca → severe growth retardation

4.6 Small for Gestational Age (SGA)

- Birth weight and/or length **< 2 SD below mean**
- Mainly from something during pregnancy (placental insufficiency, maternal illness)
- Normally catch-up growth happens in first 2 y → in SGA **catch-up fails**
- Mechanism not well understood (altered GH/IGF-1 sensitivity; GH often normal)
- May benefit from **recombinant human GH (rhGH)** → given to force growth, not to replace a missing hormone

4.7 Growth Chart Patterns

Pattern	Chart
Constitutional delay	Falls below, then late spurt → catches up to $\sim$ 50th
Familial short stature	Parallel to but below 5th throughout
Precocious puberty	Tall early → stops early → short adult
Cushing syndrome	Height falls off while weight ↑
Exogenous obesity	Weight ↑ and height ↑ (tall)

5. Growth Hormone Deficiency (GHD)

5.1 Causes

Isolated GHD (IGHD)

- Genetic:
 - **GHRH receptor mutation** → pituitary never gets the "make GH" signal
 - **GH1 gene mutation/deletion** → can't make GH
 - **GH insensitivity** (peripheral resistance) → **↑ GH but ↓ IGF-1**
- Acquired: **radiotherapy** (cranial irradiation for brain tumor / leukemia)

Multiple pituitary hormone deficiency (MPHD)

- Other hormones lost too → ACTH, TSH, LH/FSH, ADH → adrenal insufficiency + hypothyroidism
- Genetic
- Pituitary hypoplasia** + CNS/midline anomalies: **anencephaly, midfacial anomalies, cleft palate** (shared embryologic origin, Rathke's pouch)
- Acquired: trauma · infiltrative (**craniopharyngioma**) · irradiation · surgery · vascular (infarction)

5.2 Congenital Hypopituitarism

- Normal size at birth** (or 1 SD below) → fetal growth depends on insulin + IGF-2, not GH
- Grows < 25th percentile → gradually diverges
- Neonatal emergencies: **apnea, cyanosis, severe hypoglycemia**
- Micro-phallus in boys** → **additional diagnostic clue**
- ± hypoadrenalism, hypothyroidism; **prolonged neonatal jaundice** is common



Skull X-ray: craniopharyngioma

5.3 Clinical Features (Child)

- Fasting symptomatic hypoglycemia** in 10–15% (panhypopituitarism + IGHD); GH is counter-regulatory
- Intelligence usually normal**
- Proportionate** short stature; small hands and feet
- Face: **round head, short broad face, prominent frontal bone**, small nose, deep nasolabial folds, slightly bulging eyes
- Small mandible/chin; **late, crowded teeth**
- Short neck, small larynx → **high-pitched voice, stays high after puberty**
- Small genitals; delayed/absent puberty; little facial/axillary/pubertic hair; fine scalp hair

5.4 Investigations

- GH is pulsatile → random level useless → **stimulation test** (insulin / clonidine / arginine)
- Absent/low response; **peak < 10 µg/L → GHD**
- IGF-1**: normal 0.3–1.4 U/mL; stable → screening; ↑ GH + ↓ IGF-1 → insensitivity
- Other pituitary hormones; hypothalamic releasing hormones
- X-ray: skull, long bone (bone age → delayed in GHD)
- CT/MRI** → structural lesion, esp. **craniopharyngioma** (calcification on imaging)

5.5 Treatment

- Treat the underlying cause** (e.g. remove tumor)
- hGH 0.18–0.3 mg/kg/week SC, split in 5–7 doses** (mimics pulsatile release)
- GHRH
- IGF-1** → GH-insensitive patients
- Replace other hormones (MPHD)

6. Causes of Short Stature & Approach

6.1 Endocrine Causes: Hormonal Effects on Growth

Hormone	Bone age	Growth rate	Adult height
Androgen excess	Advanced	↑	↓
Androgen deficiency	Normal / delayed	Normal / ↓	Slightly ↑ / normal
Thyroxine excess	Advanced	↑	Normal / ↓
Thyroxine deficiency	Retarded	↓	↓
GH excess	Normal / advanced	↑	Excessive
GH deficiency	Retarded	↓	↓
Cortisol excess	Retarded	↓	↓
Cortisol deficiency	Normal	Normal	Normal

Adult height = effect in most treated patients

6.2 Non-endocrine Causes

- Malnutrition · chronic disease · **psychosocial (deprivation dwarfism)**
- Constitutional delay · familial short stature · syndromes
- Curves: prenatal-onset pathologic (short from birth) · postnatal-onset pathologic (falls off later) · constitutional delay · familial short stature

6.3 Assessment of a Short Child (Approach)

1) Growth chart (height, weight, HC)

- ▶ **Following** centile lines → familial, low birth weight, CDGP, syndromes, skeletal dysplasias
- ▶ **Crossing** centile lines (growth failure) → endocrine (incl. therapeutic steroids), nutrition/chronic illness, psychosocial deprivation

2) Mid-parental height → genetic target range

3) History

- ▶ Birth length, weight, HC, gestational age
- ▶ Pregnancy: infection, IUGR, drugs, alcohol/smoking
- ▶ Feeding · developmental milestones
- ▶ Family history of CDGP / other disease · consanguinity
- ▶ Chronic illness, endocrine (hypothyroidism, pituitary tumor, Cushing), psychosocial
- ▶ Medications (corticosteroids)

4) Examination

- ▶ Dysmorphic features (Turner may lack other stigmata)
- ▶ Chronic illness: Crohn, cystic fibrosis, coeliac
- ▶ Endocrine signs · disproportion (skeletal dysplasia) · pubertal stage

5) Diagnosis → **cause usually found from the above; no tests needed**

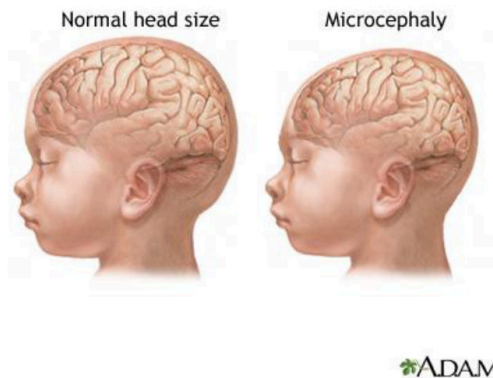
7. Tall Stature & Abnormal Head Growth

7.1 Tall Stature (homework in lecture)

Type	Details
Familial	Most common
Obesity	Puberty advanced → final height NOT increased
Secondary	Hyperthyroidism · excess sex steroids · excess GH
Syndromes	Marfan · Klinefelter · Beckwith · Sotos

7.2 Head Circumference & Microcephaly

- HC is important esp. in the **first 3 y** (maximal brain growth)
- **Microcephaly**: HC < 2 SD (~3rd percentile) for age and sex
- **Primary (genetic)** → almost always present at birth
- **Secondary (environmental)** → at birth or later after postnatal insult



Normal head vs microcephaly

ADAM.

7.3 Macrocephaly

- HC > 2 SD (~97th percentile) for age and sex
- Causes: **hydrocephalus** · enlarged brain (**megalencephaly**) · space-occupying lesions



Hydrocephalus

7.4 CSF Flow

CSF made by **choroid plexus** (lateral, 3rd, 4th ventricles)

Lateral ventricles → foramina of **Monro** → 3rd ventricle → aqueduct of **Sylvius** → 4th ventricle

- ▶ → foramina of **Luschka** + foramen of **Magendie** → subarachnoid space
- ▶ → central canal of spinal cord

Summary

Condition	Key feature	Bone age	Velocity	Final height / outcome	Management
Familial short stature	Normal variant, short parents	Normal	Low-normal	Reaches mid-parental height	Reassure
Constitutional delay	Slows in first 3–5 y, family history	Delayed (hallmark)	↓ then late spurt	Complete catch-up	Reassure
GH deficiency	Proportionate, round face, high voice, micro-phallus, hypoglycemia	Retarded	↓ (< 25th)	↓ without Tx	Treat cause; hGH 0.18–0.3 mg/kg/wk SC in 5–7 doses
SGA	Birth weight/length < –2 SD, no catch-up			Stays short	rhGH
Turner	One X; ± wide neck, wide-spaced nipples, cardiac			Short	
Down	Short by nature			Short	Own growth chart
Sanjad-Sakati	Middle East, AR, hypoparathyroidism → ↓ Ca			Severe growth retardation	
Hypothyroidism	Endocrine	Retarded	↓	↓	
Cushing / steroids	Weight ↑, height ↓	Retarded	↓	↓	
Obesity (tall)	Tall + heavy, early puberty		↑	Not increased	

Key Numbers

Item	Value	Item	Value
Short / tall stature	< 3rd / > 97th percentile (±2 SD)	Birth length / weight	~50 cm / 3–4.5 kg
Microcephaly / macrocephaly	< 3rd / > 97th (±2 SD)	1st-year length gain	~25 cm
Growth failure	Crossing > 2 percentiles	Childhood velocity	4–8 cm/y
Target height (boy)	[FH + (MH + 13)] / 2 ± 5	Peak height velocity	M ~10 · F ~8 cm/y; F 2 y earlier
Target height (girl)	[(FH – 13) + MH] / 2 ± 5	U/L ratio	1.7 birth · 1.3 3 y · 1.0 14 y · 0.9 adult
GH stimulation	Peak < 10 µg/L → GHD	IGF-1 normal	0.3–1.4 U/mL
Hypoglycemia in GHD	10–15%	Adult testis volume	20–25 mL

Ultra-Quick Differentiation (Keywords)

Keyword combination	Think of
Short, delayed bone age , family history of late bloomers, catch-up	Constitutional delay
Short, normal bone age , short parents, reaches target height	Familial short stature
Percentile 50th → 25th → 10th → 5th	Growth failure
Neonate: hypoglycemia, micro-phallus, prolonged jaundice	Congenital hypopituitarism
Round face, frontal bossing, high-pitched voice, normal IQ, proportionate	GH deficiency
↑ GH + ↓ IGF-1	GH insensitivity
Cleft palate / midline defect + hypopituitarism	Pituitary hypoplasia (MPHD)
Calcified lesion near pituitary, acquired GHD	Craniopharyngioma
Weight ↑ but height falls off	Cushing / steroids
Tall and heavy, early puberty, normal final height	Exogenous obesity

Keyword combination	Think of
Short girl, wide neck, widely spaced nipples, cardiac defect	Turner syndrome
Short, hypocalcemia, Middle East, consanguinity	Sanjad-Sakati syndrome
Birth weight < -2 SD, never catches up	SGA → rhGH
First sign of puberty (F / M)	Breast bud / testicular enlargement
Beads measuring testicular volume	Orchidometer
Tall, most common cause	Familial tall stature
Tall stature syndromes	Marfan, Klinefelter, Beckwith, Sotos
HC > 97th	Hydrocephalus, megalencephaly, mass
Main growth regulator in infancy / childhood	Nutrition / hormones
Saudi child plotted on CDC chart	Over-diagnosis of undernutrition

Case-Based Practice MCQs (answers on the next page)

1. A 13-year-old boy is below the 3rd percentile. Growth slowed in early childhood; his father was a "late bloomer". Bone age is delayed. Most likely?
 - A. Familial short stature
 - B. Constitutional delay of growth and puberty
 - C. GH deficiency
 - D. Turner syndrome
2. A short 8-year-old with short parents grows parallel to the 3rd percentile. Bone age = chronological age. Most likely?
 - A. Familial short stature
 - B. Constitutional delay
 - C. Hypothyroidism
 - D. Cushing syndrome
3. A child's height moves from the 50th (6 y) to 25th, 10th and 5th percentile by 9 y. This is:
 - A. Normal variation
 - B. Growth failure
 - C. Familial short stature
 - D. Tall stature
4. Father 176 cm, mother 160 cm. The son's target height is:
 - A. 168 cm $\pm$ 5
 - B. 174.5 cm $\pm$ 5
 - C. 161.5 cm $\pm$ 5
 - D. 181.5 cm $\pm$ 5
5. A newborn boy has severe hypoglycemia, micro-phallus and prolonged jaundice. Most likely?
 - A. SGA
 - B. Congenital hypopituitarism
 - C. Down syndrome
 - D. Hyperthyroidism
6. Which GH stimulation result suggests GH deficiency?
 - A. Peak < 10 μ g/L
 - B. Peak > 20 μ g/L
 - C. IGF-1 1.2 U/mL
 - D. High GH with high IGF-1
7. High GH with low IGF-1 indicates:
 - A. GH1 gene deletion
 - B. GH insensitivity
 - C. GHRH receptor mutation
 - D. Craniopharyngioma
8. The main regulator of growth in infancy is:
 - A. Growth hormone
 - B. Thyroid hormone
 - C. Nutritional status
 - D. Sex steroids
9. The first sign of puberty in boys is:
 - A. Pubic hair
 - B. Height spurt
 - C. Testicular enlargement
 - D. Voice change
10. Normal height velocity in childhood is:
 - A. 1–2 cm/y
 - B. 4–8 cm/y
 - C. 10–12 cm/y
 - D. 25 cm/y
11. An obese 9-year-old is tall with advanced puberty. Her final adult height will be:
 - A. Increased
 - B. Not increased
 - C. Severely reduced
 - D. Unpredictable
12. A child on long-term steroids gains weight while height falls off the chart. Bone age is:
 - A. Advanced
 - B. Retarded
 - C. Normal
 - D. Not affected by steroids
13. Which dose of hGH matches the lecture?
 - A. 0.18–0.3 mg/kg/week SC in 5–7 doses
 - B. 0.18–0.3 mg/kg/day IV
 - C. 1 mg/kg once weekly
 - D. Oral daily
14. A head circumference above the 97th percentile can be caused by all EXCEPT:
 - A. Hydrocephalus
 - B. Megalencephaly
 - C. Space-occupying lesion
 - D. Primary genetic microcephaly

Answer Key

1	B: delayed bone age = hallmark of constitutional delay; family history; complete catch-up
2	A: normal variant, normal bone age, reaches mid-parental height
3	B: crossing > 2 percentiles = growth failure
4	B: $[176 + (160 + 13)] / 2 = 174.5 \text{ cm} \pm 5$
5	B: neonatal hypoglycemia + micro-phallus + prolonged jaundice
6	A: stimulated peak < 10 µg/L suggests GHD
7	B: receptors don't respond → GH high, IGF-1 low
8	C: nutrition in infancy; hormones in childhood
9	C: testicular enlargement (girls: breast development)
10	B: 4–8 cm/year
11	B: obesity advances puberty; final height not increased
12	B: cortisol excess → retarded bone age, ↓ growth, ↓ adult height
13	A: 0.18–0.3 mg/kg/week SC split into 5–7 doses
14	D: macrocephaly causes = hydrocephalus, megalencephaly, space-occupying lesions